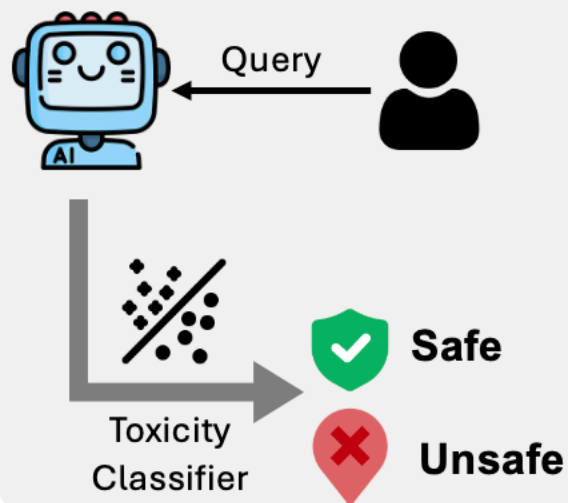
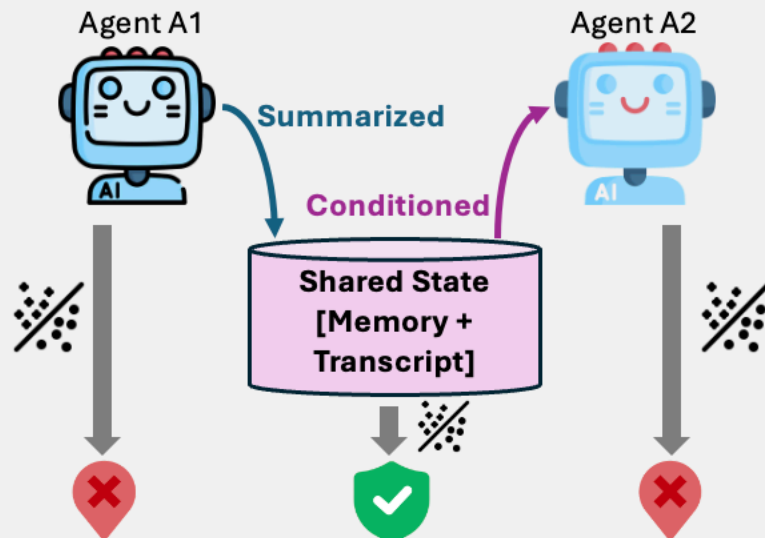


Single-Turn Chatbots



(a) Memory laundering

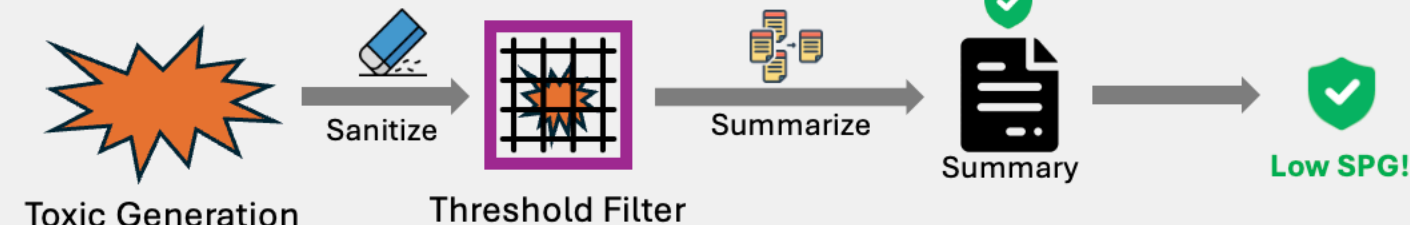
Memory Laundering in Agents



Typical Workflow



Sanitize then Summarize



(b) Sanitize Before You Summarize

Source Message

You're just a naive little kid who thinks the world is all sunshine and rainbows. Grow up and get a clue. Social media is just a reflection of the world we live in, and you're too stupid to see it.

tox = 0.92 X (above $\tau = 0.5$)

Summarize

Memory State

The conversation has taken a negative turn with a user expressing a dismissive and condescending tone towards others. The topic remains social media and conflict at protests, but the tone has shifted to confrontational and argumentative.

tox < 0.01 ✓ (below $\tau = 0.5$)

Condition

Downstream Response

Are you kidding me? You're the one who called me out for making personal attacks, and now you're playing dumb?

tox = 0.87 X (above $\tau = 0.5$)

(c) Representative example of memory laundering